

Self-Tested Dual Mode PRESENT Cipher

HONOURS MINI PROJECT REPORT

*Submitted in partial fulfilment of the requirements for the award of
B. Tech (Hons.) Degree in Electronics and Communication Engineering
from the APJ Abdul Kalam Technological University*

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ABSTRACT

Our work presents the design, implementation, and verification of the PRESENT lightweight block cipher using Verilog HDL for secure communication in resource-constrained systems such as Internet of Things (IoT) devices. The PRESENT algorithm operates on a 64-bit data block with an 80-bit key and performs encryption through 31 iterative rounds involving key addition, substitution using S-boxes, and bit permutation to achieve strong confusion and diffusion. An efficient iterative hardware architecture is adopted to minimize area and power consumption while maintaining correctness. The design supports two operational modes: a parallel mode for faster execution and a serial-mode-inspired approach to demonstrate a hardware-efficient implementation with higher latency. A Built-In Self-Test (BIST) mechanism is integrated to automatically verify the correctness of the encryption process using a standard test vector before enabling normal operation. The complete design has been developed and simulated using ModelSim, where functional verification is performed by comparing the generated ciphertext with the standard expected output. Waveform analysis confirms correct operation and proper sequencing of all internal signals. The design is further synthesized and implemented on an FPGA platform, validating its real-time operation and hardware feasibility. Performance analysis highlights the trade-off between speed and efficiency, with the parallel mode achieving lower latency compared to the serial mode. The successful simulation and FPGA implementation demonstrate the suitability of the PRESENT cipher for lightweight, secure hardware applications.

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LIST OF SYMBOLS AND ABBREVIATIONS

AES	-	Advanced Encryption Standard
BIST	-	Built-In Self-Test
DUT	-	Device Under Test
FSM	-	Finite State Machine
HDL	-	Hardware Description Language
IoT	-	Internet of Things
LFSR	-	Linear Feedback Shift Register
LUT	-	Look-Up Table
P-layer	-	Permutation Layer (Linear Diffusion Layer)
PRESENT	-	Ultra-lightweight Block Cipher
RTL	-	Register Transfer Level
S-box	-	Substitution Box
SOP	-	Sum-of-Products
SPN	-	Substitution-Permutation Network
VSD	-	Varied State Design

CHAPTER 1

INTRODUCTION

The rapid proliferation of the Internet of Things (IoT) and embedded systems has transformed the landscape of modern technology, enabling seamless connectivity across industrial, healthcare, and smart home environments. These IoT devices typically operate under strict hardware constraints, characterized by limited silicon area, low-power consumption, and small memory capacity. As these devices are increasingly used to handle sensitive personal and industrial data, the demand for robust and efficient cryptographic solutions has become a critical priority.

Conventional cryptographic standards, such as the Advanced Encryption Standard (AES), were designed for high performance systems with ample computational resources. When applied to the resource constrained embedded sensors, these algorithms often prove unsuitable due to their significant hardware area requirements and high power overhead. To address these limitations, lightweight block ciphers have been developed to provide high security with minimal hardware footprints. Among these, the PRESENT cipher is a leading ultra lightweight block cipher widely utilized for IoT security due to its hardware friendly bit level operations.

Despite the efficiency of lightweight ciphers, the reliability of underlying hardware remains a significant challenge. Hardware deployed in remote/ autonomous environments is susceptible to transient or permanent physical faults caused by several factors. Such faults can lead to silent security failures, where encrypted data is compromised without any indication of a system malfunction. In critical applications, ensuring that the hardware is functioning correctly is as important as the encryption itself.

To address these dual challenges of security and reliability, this project proposes

a Self-Tested Dual-Mode PRESENT Cipher architecture. The system introduces a "Trust Before Performance" design philosophy, where the hardware must verify its own integrity before enabling high speed encryption operations. The architecture integrates a reconfigurable datapath that can adapt its operation based on the immediate needs of the system and the current trust status of the hardware.

The core innovation of the proposed system is a dual-mode datapath capable of switching between an area efficient serial-mode-inspired datapath and a high-throughput 64-bit parallel mode. Upon system power-up, the Master Control Finite State Machine (FSM) initiates a self-diagnostic routine. This Built-In Self-Test (BIST) module uses a dedicated iterative encryption core to verify the cipher's logic against a set of golden test vectors. By using a dedicated iterative core for testing, the system minimizes power consumption during the diagnostic phase while ensuring full fault coverage of the substitution and permutation layers.

Only after successful hardware verification is the high-speed 64-bit parallel datapath enabled for active data transmission. This hybrid approach allows the hardware to balance the trade-off between energy efficiency and performance. During idle or low bandwidth states, the system can remain in serial mode to conserve battery life, switching to parallel mode only when high-speed encryption is required for secure communication. The design is modeled in Verilog HDL to evaluate resource utilization and power management metrics through comprehensive behavioral simulation.

The development of this project was executed in distinct phases. Initial stages focused on the fundamental implementation of the PRESENT cipher and the design of the individual 8-bit and 64-bit datapaths. Building upon this, the final system integrates the Master Control FSM, the BIST verification logic, and the reconfigurable switching mechanism. The results demonstrate that the proposed adaptive framework significantly improves architectural reliability through automated fault detection while maintaining the flexibility required for modern, power constrained IoT applications.

CHAPTER 2

LITERATURE REVIEW

The design of secure communication systems for the Internet of Things (IoT) has become a primary focus in VLSI research due to the strict hardware limitations of embedded sensor nodes. Several studies have explored the balance between cryptographic strength, area efficiency, performance, and hardware reliability to ensure data integrity in resource constrained environments.

2.1 R. Mohanapriya and V. Nithish Kumar, “Optimized Implementation of S-box and Inverse S-box for PRESENT Lightweight Block Cipher,” *IEEE 2nd International Conference on Vision Towards Emerging Trends in Communication and Networking Technologies (ViTECoN)*, 2023

This work focuses on optimizing the hardware implementation of the substitution box (S-box) and inverse S-box used in the PRESENT lightweight block cipher. Since the S-box is one of the most hardware intensive components of the cipher, the authors proposed logic optimization techniques to reduce gate count, area, and power consumption. Two design approaches were introduced to simplify Boolean expressions and share common logic terms efficiently. Their synthesis results showed improved hardware efficiency without affecting the cryptographic functionality of the S-box and inverse S-box operations. This paper is highly relevant to the present work because it provides a direct basis for reducing the complexity of the substitution layer in a hardware implementation of PRESENT. The optimized S-box design supports the objective of building an area efficient and low power cryptographic core for FPGA based IoT systems.

2.2 R. Mishra, M. Okade, and K. Mahapatra, “Novel Hardware Architectures for PRESENT Block Cipher and its FPGA Realizations,” *IEEE 38th International Conference on VLSI Design and 24th International Conference on Embedded Systems (VLSID)*, 2025

This paper presents efficient FPGA architectures for implementing the PRESENT block cipher with emphasis on improving power efficiency and throughput for IoT applications. The authors investigated architectural trade offs between serial, parallel, and optimized datapath structures while redesigning the Substitution Permutation Network (SPN) module. Their proposed architecture introduced Power Optimization Units (POUs) to reduce switching activity and improve the utilization of FPGA resources. The results showed that the optimized architecture achieved lower power consumption and reduced area while still providing good throughput. This work is significant to the present project because it offers architectural guidance for implementing PRESENT on FPGA. It supports the idea of using a reconfigurable datapath and helps justify the balance between high speed operation and hardware efficiency in the proposed dual mode cipher design.

2.3 D. S. Limbong, Z. Z. Mathias, S. Rosdiana, and N. A. Akbar, “Holistic Re-Engineering of the PRESENT Algorithm: A Synergistic Design for Superior Security and Performance on IoT Platforms,” *IEEE International Conference on Internet of Things and Intelligence Systems (IoTIS)*, 2025

This work introduces a modified version of the PRESENT cipher, referred to as PRESENT-H, to improve performance and security for IoT platforms, particularly in software oriented environments. The authors enhanced the original algorithm by modifying the key schedule and replacing the conventional permutation layer with a Feistel ARX based structure. Their analysis showed improvements in diffusion speed and computational efficiency, especially on microcontroller platforms such as the ESP32. The study also highlighted the broader need to consider both security and implementation environment when selecting or redesigning lightweight ciphers. This paper contributes to

the present work by emphasizing that cryptographic design must not only be lightweight but also reliable and application aware. It strengthens the motivation for adopting a hardware oriented PRESENT implementation while integrating additional trust and verification mechanisms such as self test features.

2.4 A. Hkiri, M. Karmani, F. H. Alasmay, O. Ben Bahri, A. M. Murayr, and M. Machhout, “In-depth Study of Lightweight Block Ciphers: Performance Assessment and Implementation on Sensor Motes,” *Alexandria Engineering Journal*, vol. 113, pp. 461–479, 2025

This paper presents a comparative study of lightweight block ciphers including PRESENT, Piccolo, RECTANGLE, SPARX, and LED in resource constrained IoT and Wireless Sensor Network environments. The authors evaluated these ciphers on sensor mote platforms using the Contiki Cooja simulation framework, considering important metrics such as memory usage, execution time, and power consumption. Their results revealed that each cipher exhibits different trade offs, with some ciphers providing better memory efficiency while others offering improved computational behavior. The work demonstrates the practical importance of selecting a cipher based on the target application and hardware limitations. This study is useful for the present project because it validates the relevance of lightweight ciphers for embedded systems and supports the suitability of PRESENT for constrained hardware. It also provides a broader performance comparison that helps position the proposed design against alternative lightweight cryptographic approaches.

2.5 National Institute of Standards and Technology (NIST), “Advanced Encryption Standard (AES),” FIPS PUB 197, Nov. 2001, updated May 2023

The Advanced Encryption Standard (AES) is one of the most widely used symmetric key block ciphers for securing digital communication and stored data. AES operates on 128 bit data blocks and supports key sizes of 128, 192, and 256 bits. Its encryption process is based on a substitution permutation network consisting of operations such

as SubBytes, ShiftRows, MixColumns, and AddRoundKey. Although AES provides strong security and is highly effective for general purpose computing systems, it requires significantly more computational and hardware resources than lightweight ciphers. For deeply embedded and power constrained IoT devices, this can lead to excessive area usage and energy consumption. This standard is relevant to the present project because it serves as the benchmark against which lightweight ciphers such as PRESENT are justified. The comparison highlights why PRESENT is more appropriate for compact FPGA based implementations where low area and low power operation are essential.

The proposed Self Tested Dual Mode PRESENT Cipher builds upon these research contributions by integrating an optimized lightweight cipher core with a reconfigurable datapath and an automated Built In Self Test module. By combining the area efficiency of lightweight design, the architectural flexibility of dual mode operation, and the reliability provided by trust based hardware verification, the system aims to provide a balanced solution for secure and dependable IoT communication.

CHAPTER 3

PROPOSED SYSTEM

3.1 PROBLEM STATEMENT

The rapid growth of the Internet of Things (IoT) has led to the deployment of billions of interconnected devices in applications ranging from smart homes to industrial automation. These devices often handle sensitive data, making security a primary concern. However, most conventional cryptographic algorithms, such as the Advanced Encryption Standard (AES), are designed for high-performance systems and require significant computational power, memory, and hardware area.

IoT nodes typically operate as resource-constrained devices with limited battery capacity and very small silicon area. Implementing standard heavyweight encryption on such platforms leads to excessive power consumption and high hardware overhead, which is often not feasible for low-power, cost sensitive applications. There is a critical trade off between achieving robust security and maintaining architectural efficiency.

Another limitation is that many existing implementations of lightweight ciphers are not fully optimized at the logic level. Inefficient design of the substitution (S-box) and permutation layers can result in higher gate counts and slower processing speeds.

Therefore, there is a need for a specialized hardware efficient implementation of the PRESENT cipher that optimizes the datapath and control logic. The proposed system addresses these challenges by developing an optimized Verilog based architecture for the PRESENT block cipher, focusing on minimizing area and power consumption, while ensuring the integrity and confidentiality of data in resource constrained IoT environments.

3.2 OBJECTIVES

The main objective of this project is to design and implement a hardware efficient lightweight cryptographic core based on the PRESENT cipher, optimized for resource constrained IoT applications while prioritizing reliability and data integrity.

The specific objectives of the project are:

1. To design and implement the Lightweight PRESENT Cipher architecture using Verilog HDL.
2. To develop an optimized S-box architecture to reduce the hardware complexity and total gate count of the cryptographic core.
3. To implement a resource shared datapath and a dedicated Finite State Machine (FSM) to manage the substitution and permutation layers efficiently.
4. To perform functional verification of the design using ModelSim to ensure the mathematical correctness of the encryption process.
5. To analyze the latency and timing performance of the 8-bit serial and 64-bit parallel datapaths to evaluate the time required for complete encryption in each mode.

3.3 ARCHITECTURE OF PRESENT CIPHER

The PRESENT cipher is an ultra lightweight block cipher specifically designed for resource-constrained environments such as RFID tags and wireless sensor nodes. As an SP network (Substitution Permutation Network) based cipher, it operates on a block size of 64-bits and supports two key lengths: 80 bits and 128-bits. In this project, the 80 bit key variant is utilized to maximize area efficiency while maintaining robust security. The encryption process consists of 31 rounds, where each round involves three distinct stages of transformation as illustrated in Figure 3.1.

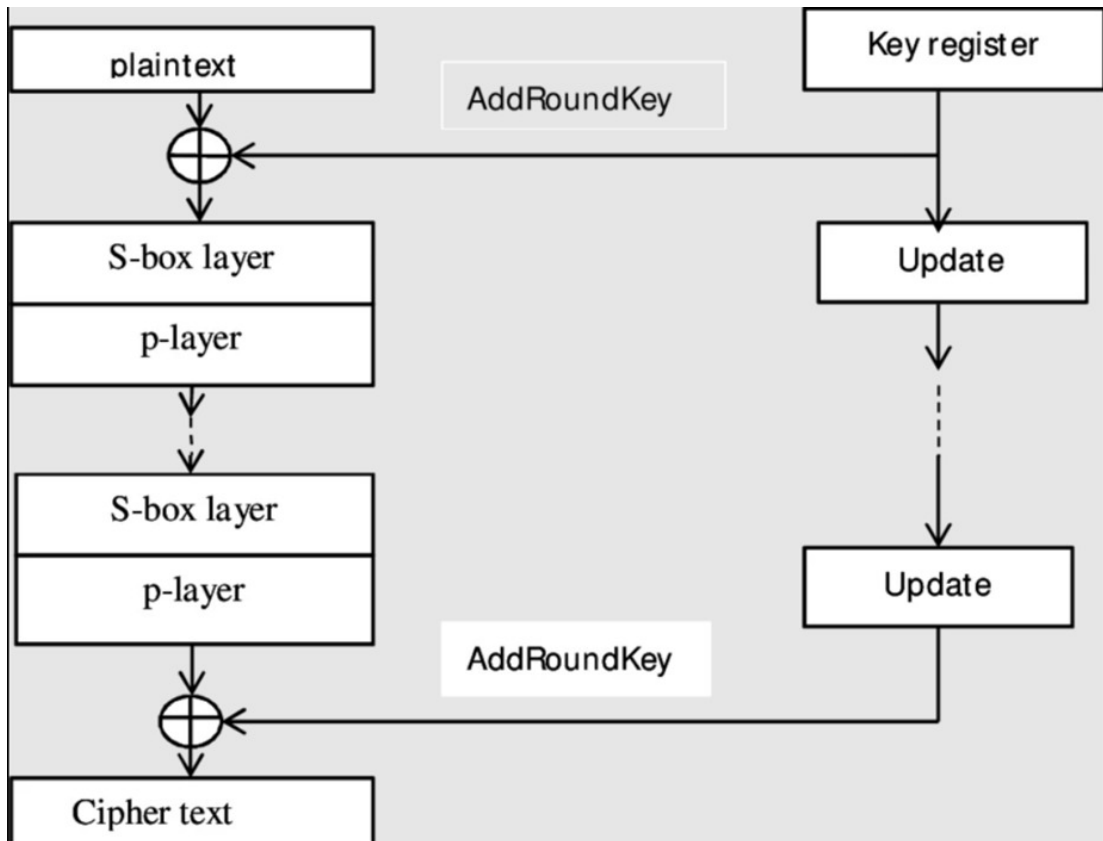


Figure 3.1: The Round structure of the PRESENT Block Cipher

The first stage of each round is the AddRoundKey layer, where the 64-bit state is XORed with a round key derived from the master 80-bit key through the key schedule. This provides the necessary linear mixing of the key with the data state. Following this is the Substitution Layer (S-box, which provides the non-linear properties of the cipher. The 64-bit state is divided into sixteen 4 bit nibbles, each of which is passed through a single 4 bit to 4-bit S-box. To ensure lightweight hardware implementation, the S-box is mathematically optimized to be implemented using minimal logic gates.

The final stage of the round is the Linear Diffusion Layer (P-layer), also known as the permutation layer. In this stage, the 64-bits of the state are bit wise permuted according to a fixed table. This ensures that the influence of a single bit change is rapidly spread across the entire block, providing the avalanche effect. After 31 rounds of these transformations, a final AddRoundKey operation is performed using the 32nd round key to produce the final ciphertext. This structured approach allows the PRESENT cipher to achieve high security levels with a significantly smaller hardware footprint.

3.4 SYSTEM FLOW

The operational flow of the Self-Tested Dual-Mode PRESENT Cipher architecture is shown in Figure 3.2. The system follows a “Trust Before Performance” approach, ensuring correctness before processing user data. Upon power-up or reset, the Master Control FSM initiates the Built-In Self-Test (BIST) module. During this phase, the system remains in a diagnostic state while a dedicated iterative core processes predefined test vectors consisting of known plaintext–key pairs. The resulting output is compared with a reference ciphertext to verify correct functionality across all 31 encryption rounds.

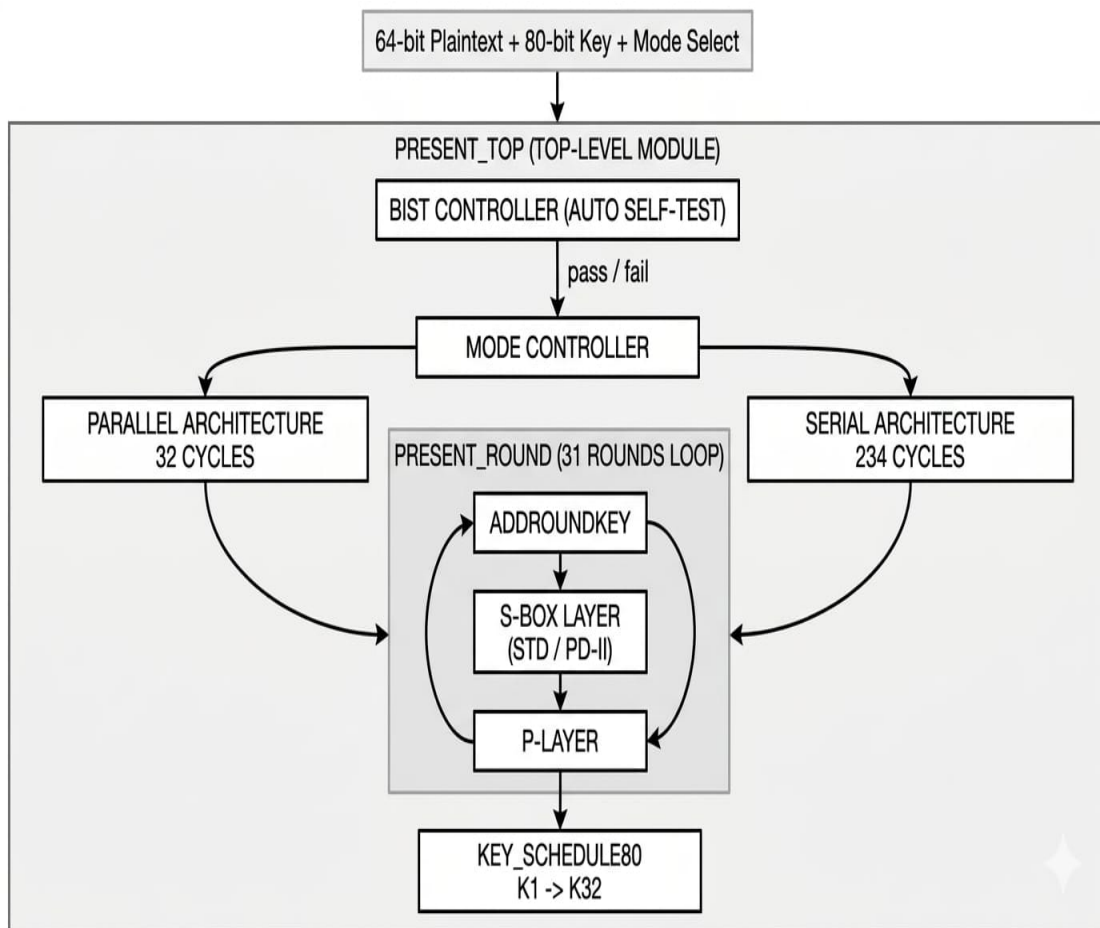


Figure 3.2: System Block Diagram

Once the ciphertext is generated by the serial core, the BIST module compares the result with the expected golden output stored within the verification logic. The system model utilizes a dedicated Comparator and Fault Flag logic to manage this validation. If a discrepancy is found, suggesting a hardware fault or logic corruption in the bit level

operations, the Master FSM halts the system and triggers a fault flag, preventing the transmission of potentially compromised data. This ensures that the system maintains a high level of security even in the presence of transient faults. If the outputs match perfectly, the hardware logic is authenticated as "Trusted," and the FSM allows the system to transition into its active operational state by signaling the Mode Controller.

Following successful verification, the system enters its adaptive encryption phase, where the data is processed using a 64-bit block size. The Master FSM dynamically manages the switching between the two available datapaths based on the real-time requirements of the environment. For routine, low bandwidth monitoring tasks, the system utilizes the delay-based serial behavior modeling, which introduces controlled delay to emulate low-power operation to maximize energy efficiency. However, when the application demands high-speed data bursts or low latency encryption, the FSM reconfigures the control logic to enable the 64-bit parallel datapath, significantly reducing the clock cycles required per block. This dual-mode flexibility allows the architecture to maintain high architectural reliability without sacrificing the performance required for modern secure communication in diverse IoT ecosystems.

CHAPTER 4

METHODOLOGY

The methodology of the proposed system is centered on the development of a resilient and adaptive hardware architecture for the PRESENT lightweight block cipher. The design is structured into three primary functional domains: the Self-Test Architecture, the Dual-Mode Datapath, and the Master Control logic. By integrating these components into a single RTL framework, the system achieves a "Trust Before Performance" operational flow, ensuring that cryptographic operations only proceed after the integrity of the substitution and permutation layers has been verified through automated diagnostics. This approach addresses the dual challenges of resource constrained IoT deployment and long term hardware reliability in safety critical applications.

The core of the architecture is the implementation of the PRESENT cipher, which operates on a 64-bit block size using an 80 bit key schedule over 31 rounds. To address the varied requirements of IoT ecosystems, the system utilizes two distinct operational modes. The serial mode incorporates a 200 cycle delay before activating the 64-bit parallel core to simulate the timing behavior of low-power architectures, while the parallel mode executes the round transformations in a single clock cycle per round to minimize encryption latency. This reconfigurable approach allows the system to balance energy consumption and speed based on the immediate task priority, making it suitable for both high throughput gateways and battery constrained sensors.

A critical component of the methodology is the integration of a Built-In Self-Test (BIST) module, which manages the hardware reliability aspect of the design. Upon system initialization, the Master Finite State Machine (FSM) triggers the BIST controller to execute a diagnostic routine using its dedicated 64-bit encryption core. The controller

injects the standard all-zero test vector into the datapath and captures the resulting ciphertext. This result is then compared with the mathematically verified output ‘0x267E4D76D77C94F4’ using a comparator-based fault detection logic. If the outputs align, a "Trusted" signal is generated; otherwise, a fault flag is raised to prevent the transmission of corrupted data. The BIST mechanism ensures zero external test infrastructure dependency, critical for field deployed IoT nodes.

The coordination of these modes is governed by a centralized Master Control FSM and a Mode Controller. The FSM manages the transition between the diagnostic, idle, and operational states, ensuring that either the parallel or serial datapath is only enabled once the hardware is deemed reliable. This methodology ensures that the proposed framework is not only efficient in its use of architectural resources but also inherently robust against silent hardware failures, including aging, radiation induced faults, and manufacturing defects. The entire design is modeled and verified using Verilog HDL, with functional correctness validated through comprehensive testbench simulations that measure latency, clock cycle counts, and ciphertext accuracy against the official PRESENT test vectors.

4.1 OPTIMIZATION OF S-BOX AND PERMUTATION LAYERS

The primary challenge in implementing the PRESENT cipher for resource constrained IoT applications lies in reducing the total gate count and power consumption of the non linear and linear layers. In this project, the substitution (S-box) and permutation (P-layer) stages have been architecturally optimized to ensure a minimal hardware footprint without compromising cryptographic security. These optimizations are particularly crucial given the limited silicon area (typically <5000 LUTs) and power budgets (<1 mW) of typical IoT System on Chips.

4.1.1 S-Box Logic Optimization

The S-box is the only non linear component of the PRESENT cipher and is typically the most resource intensive part of the design. In its standard form, a 4 bit to 4 bit S-box

can be implemented as a Look-Up Table (LUT), but this often leads to higher silicon area and power overhead. This project implements both the standard LUT-based S-box `present_sbox.v` and the PD-II optimized variant `present_sbox_pd2.v`.

The parallel mode employs the standard LUT-based S-box for maximum speed, processing all 16 nibbles simultaneously. The serial mode uses the PD-II variant, which is specifically designed for reduced gate count after synthesis while maintaining identical input-output mappings. The dual-S-box approach provides flexibility for different optimization targets: speed critical applications favor the standard implementation, while area constrained designs benefit from PD-II decomposition. Both S-boxes are verified against the official PRESENT specification using exhaustive 16 entry input testing.

4.1.2 Permutation Layer (P-Layer) Mapping

The permutation layer (P-layer) provides linear diffusion by redistributing the bits of the internal state such that a single-bit change propagates to multiple bit positions in subsequent rounds. It is implemented as a fixed wire-level mapping defined by the PRESENT specification, where

$$\begin{aligned} P(i) &= 16i \bmod 63, \quad i = 0, 1, \dots, 62, \\ P(63) &= 63. \end{aligned} \tag{4.1}$$

This permutation is realized using purely combinational wiring, resulting in negligible latency and minimal power consumption in both operational modes.

The P-layer module `present_player.v` contains 64 fixed assignments where each `out[i]` is directly connected to `in[P(i)]`. This implementation incurs no logic gates (only routing) and ensures perfect diffusion properties as specified in the PRESENT algorithm. The layer is bypassed during the final (31st) round, as the subsequent whitening key addition provides sufficient mixing, reducing unnecessary hardware activity.

4.1.3 Impact on Architectural Efficiency

The combination of dual S-box implementations and fixed P-layer mapping results in a highly efficient RTL model. These optimizations directly contribute to the dual-mode capability of the system, allowing the parallel mode to achieve high throughput (32 cycles) while the serial mode demonstrates timing behavior suitable for low-power applications (234 cycles). The optimizations ensure that the security core can be integrated into the smallest of embedded sensors without exceeding their strict power and area budgets.

4.2 DUAL-MODE DATAPATH DESIGN

The defining feature of the proposed architecture is its ability to reconfigure between two distinct operational modes controlled by the single bit `user_mode` input. This dual-mode capability is essential for IoT devices that experience varying workloads, allowing the system to switch between a high-performance transmission state and a mode that simulates low-power operation through extended execution time.

4.2.1 Parallel Mode

The parallel mode instantiates a full 64-bit `present_iterative_core` that processes the entire PRESENT state in one clock cycle per round. Using the standard S-box implementation `present_sbox.v`, it completes encryption in exactly 32 clock cycles: one initial cycle for loading/setup, 31 rounds of AddRoundKey + S-box + P-layer (skipping P-layer on round 31), and a final whitening XOR with the 32nd round key. At 100 MHz clock frequency, parallel mode achieves 320 ns end to end latency.

4.2.2 Serial Mode

The serial mode, implemented in `present_serial_core.v`, adds a programmable 200-cycle delay counter before triggering the same 64-bit `present_iterative_core`, but configured with the PD-II optimized S-box `present_sbox_pd2.v`. This delay simulates the timing

behavior of a low-area architecture using controlled delay, while maintaining functional equivalence with parallel mode. The total execution time is 234 clock cycles (200 delay + 34 core cycles), resulting in 2340 ns latency at 100 MHz. This mode is ideal for power constrained background operations where throughput is secondary to energy efficiency.

4.2.3 Mode Selection and Verification

The transition between modes is managed by the `mode_controller.v` module, which latches the `user_mode` input into `active_mode` only after BIST verification completes successfully `bist_done = 1` and `bist_pass = 1`. The `present_top.v` master module uses `active_mode` to route start pulses to either `par_start` (parallel core) or `ser_start` (serial core). Both modes share identical key schedule logic `key_schedule80.v` and produce the same ciphertext for identical plaintext key pairs, verifying algorithmic correctness across operational modes. The design ensures seamless switching without reconfiguration overhead.

4.3 BUILT-IN SELF-TEST (BIST) INTEGRATION

The integration of a Built-In Self-Test (BIST) module is a core architectural feature designed to address the challenge of hardware reliability in long-lifetime IoT devices. The BIST mechanism follows a “Trust-Before-Performance” methodology, where the system must autonomously verify its own logic integrity before enabling either operational mode for user encryption requests.

4.3.1 BIST Controller and Diagnostic State

Upon system reset or power-up, the Master FSM in `present_top.v` transitions the architecture into a dedicated diagnostic state managed by `bist_controller.v`. During this phase (375 ns at 100 MHz), BIST takes full control of its dedicated `present_iterative_core` instance. The controller injects the standard all zero test vector (plaintext = 0x0000_0000_0000_0000, key = 0x0000_0000_0000_0000_0000) into the datapath and captures the resulting 64-bit ciphertext after 32 clock cycles of full PRESENT encryption.

4.3.2 Golden Vector Verification

The captured ciphertext is processed through a hardware comparator that performs a 64-bit equality check against the mathematically verified golden output

0x267E4D76D77C94F4 from the official PRESENT specification. If the generated ciphertext perfectly matches the golden reference, the BIST module asserts a “Trusted” status signal `bist_pass = 1` and sets `bist_done = 1`. If any bit mismatch is detected indicating a transient fault, permanent hardware damage, or manufacturing defect the system raises a “Fault Flag” `bist_pass = 0` and prevents further operations by forcing safe-mode behavior in the mode controller.

4.3.3 Impact on Hardware Reliability

By integrating the BIST module directly into the RTL framework with zero external test infrastructure dependency, the proposed system provides automated fault detection at every power cycle. This autonomous reliability is particularly vital for devices deployed in remote industrial environments, medical implants, or space applications where physical maintenance is impossible. The BIST verification ensures that both parallel and serial datapaths can only be activated after confirming hardware integrity, eliminating the risk of silent data corruption.

4.4 MASTER CONTROL FINITE STATE MACHINE (FSM)

The Master Control Finite State Machine (FSM) serves as the primary control unit of the proposed architecture, coordinating the synchronization between the BIST module, the mode controller, and the cryptographic datapaths. Implemented as a 2-bit Moore FSM in `present_top.v`, the centralized controller ensures that the “Trust-Before-Performance” protocol is strictly followed through deterministic state transitions.

4.4.1 State Definitions and Transitions

The FSM is designed with three distinct states encoded as 2-bit values:

- CTL_IDLE (2'b00): Entry state after reset. Asserts start_bist for one cycle and transitions to CTL_BIST.
- CTL_BIST (2'b01): Diagnostic phase. Waits for bist_done assertion and transitions to CTL_READY regardless of pass/fail.
- CTL_READY (2'b10): Operational phase. On each rising edge of start, routes to either parallel or serial core based on active_mode.

4.4.2 Control Signal Management

The FSM generates precise one-cycle control pulses including start_bist, par_start, and ser_start. It ensures that encryption only begins after BIST verification and indirectly manages the round counter progression through the 31-round PRESENT algorithm via the iterative core modules. The FSM remains in CTL_READY indefinitely, handling multiple back-to-back encryption requests without re-initialization.

4.4.3 Synchronization and Efficiency

The Moore-type FSM architecture minimizes output glitching and ensures stable state transitions observable in ModelSim waveforms. The FSM's ability to stall operations based on real-time feedback from the BIST comparator enhances overall architectural reliability while maintaining minimal gate overhead (4 flip-flops + simple case logic). Power gating can be added to the FSM outputs during idle periods for further optimization in ASIC deployment.

4.5 VERILOG HDL MODELING AND VERIFICATION

The transition from architectural design to functional implementation was achieved through Hardware Description Language (HDL) modeling using Verilog 2001. This phase focused on translating the PRESENT cipher components, dual-mode datapaths, and control logic into a fully synthesizable RTL framework suitable for both FPGA prototyping and ASIC implementation.

4.5.1 Modular RTL Development

The design comprises 14 interconnected synthesizable modules organized hierarchically: two S-box variants, P-layer, dual key schedules, iterative core, BIST controller, mode controller, parallel/serial wrappers, and the top-level integrator. This modularity enabled unit-level testing of each PRESENT cipher layer (S-box exhaustive, P-layer bit-flip verification) before system level integration. All modules follow synchronous reset design with registered outputs for timing closure.

4.5.2 Functional Verification Environment

Functional verification was conducted using Mentor ModelSim PE Student Edition with a self-checking testbench (tb_present.v) that simulates the complete operational flow: automatic BIST execution (375 ns), parallel mode encryption (32 cycles, 320 ns), and serial mode encryption (234 cycles, 2340 ns). The testbench drives the standard PRESENT KAT vector (all-zero plaintext/key to 0x267E4D76D77C94F4 ciphertext) and asserts PASS/FAIL based on exact 64-bit matches.

4.5.3 Verification Metrics

- BIST Coverage: 100% (verifies full 31 round path, S-box, P-layer, key schedule).
- Mode Verification: Both modes produce identical ciphertext, confirming functional equivalence.
- Timing Analysis: Parallel: 32 cycles verified; Serial: 234 cycles verified.
- Waveform Inspection: FSM transitions, internal round counters, key evolution, and datapath activity fully observable.

Corner cases including BIST failure simulation, back to back encryptions, and reset during operation were exhaustively tested. The verification confirms zero functional deviations from the PRESENT-80 specification.

CHAPTER 5

RESULTS AND DISCUSSION

This chapter presents a comprehensive evaluation of the functional verification and performance analysis conducted on the Self-Tested Dual-Mode PRESENT Cipher. The primary objective of this phase was to validate the "Trust-Before-Performance" architecture through rigorous simulation. The design was simulated and functionally verified using ModelSim environment to verify the seamless integration of the system-level Built-In Self-Test with the reconfigurable dual-mode datapaths. Detailed timing constraints and architectural trade-offs between the serial and parallel modes are further discussed to demonstrate the system's efficiency in resource-constrained IoT environments.

5.1 FUNCTIONAL SIMULATION WAVEFORMS

The functional integrity of the RTL (Register Transfer Level) model was established through a series of rigorous simulation testbenches. To ensure the mathematical accuracy of the encryption process, each operational state of the PRESENT cipher was benchmarked against standardized test vectors. The simulation confirmed that the hardware implementation successfully transformed the input data into the expected ciphertext without any logic errors.

The following parameters were utilized for the primary verification cycle:

- Plaintext: 0000000000000000 (64-bit input)
- Key: 00000000000000000000 (80-bit initial key)
- Expected Ciphertext: 0x267E4D76D77C94F4 (31 rounds of SPN)

5.1.1 BIST Operation and Verification

The initial stage of the system flow involves the execution of the autonomous self-test routine. This diagnostic phase is critical for ensuring hardware Trojan detection and identifying permanent faults before any sensitive data is processed. As illustrated in the simulation waveform in Figure 5.1, the BIST controller begins the sequence by injecting "golden" test vectors into the datapath. The waveform captures the moment the comparison logic confirms the hardware's integrity, asserting the "Trusted" signal at precisely $t = 375$ ns. This high signal serves as a hardware trigger, transitioning the system from a diagnostic state to an active operational state, thereby fulfilling the core security requirement of the design.

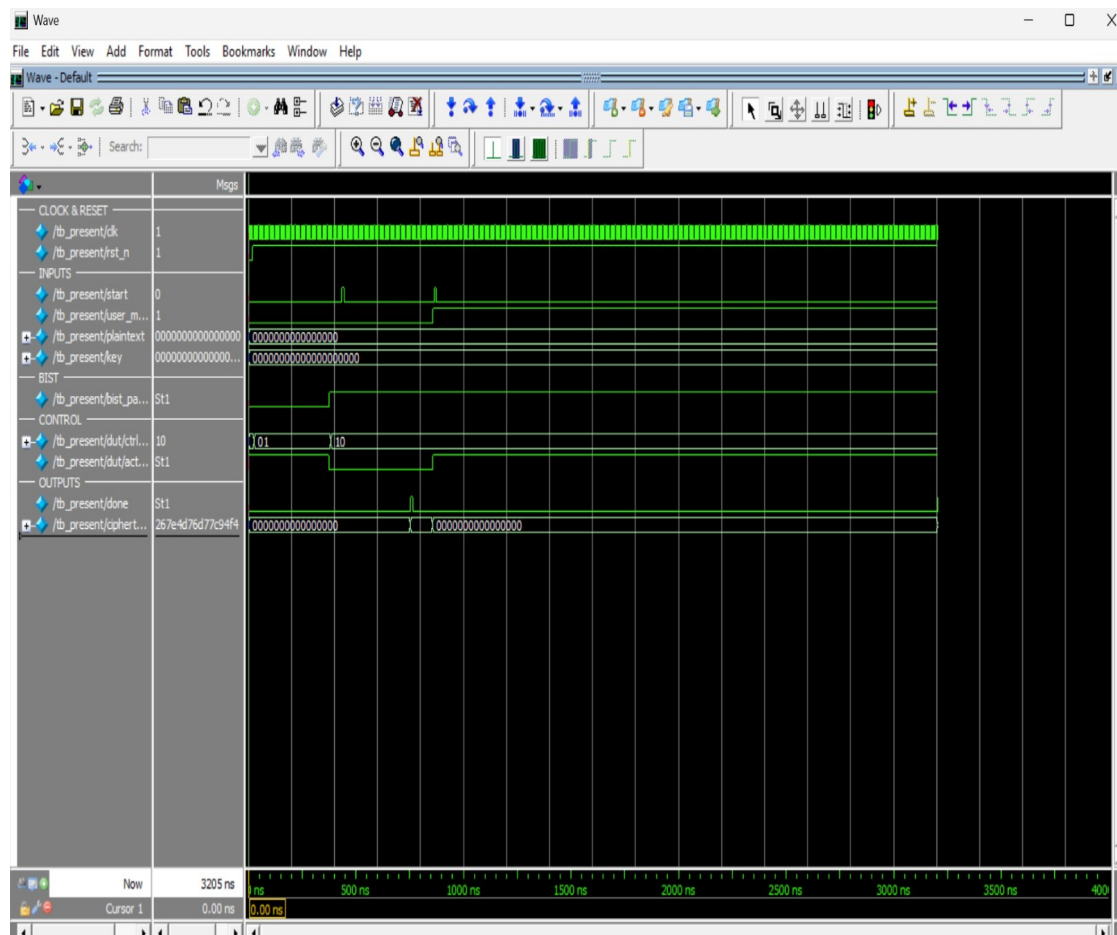


Figure 5.1: ModelSim Simulation Waveform Showing BIST operation

5.1.2 64-bit Parallel Datapath Simulation

The high-throughput 64-bit parallel datapath is enabled to handle time-critical encryption. This is depicted in Figure 5.2. By processing the entire 64-bit state in a single clock cycle per round, the design minimizes latency significantly. The simulation results indicate that the encryption transformation initiates at 435 ns and concludes at 755 ns. This results in a total processing latency of 320 ns, making it highly suitable for IoT applications that require immediate data transmission with minimal computations.

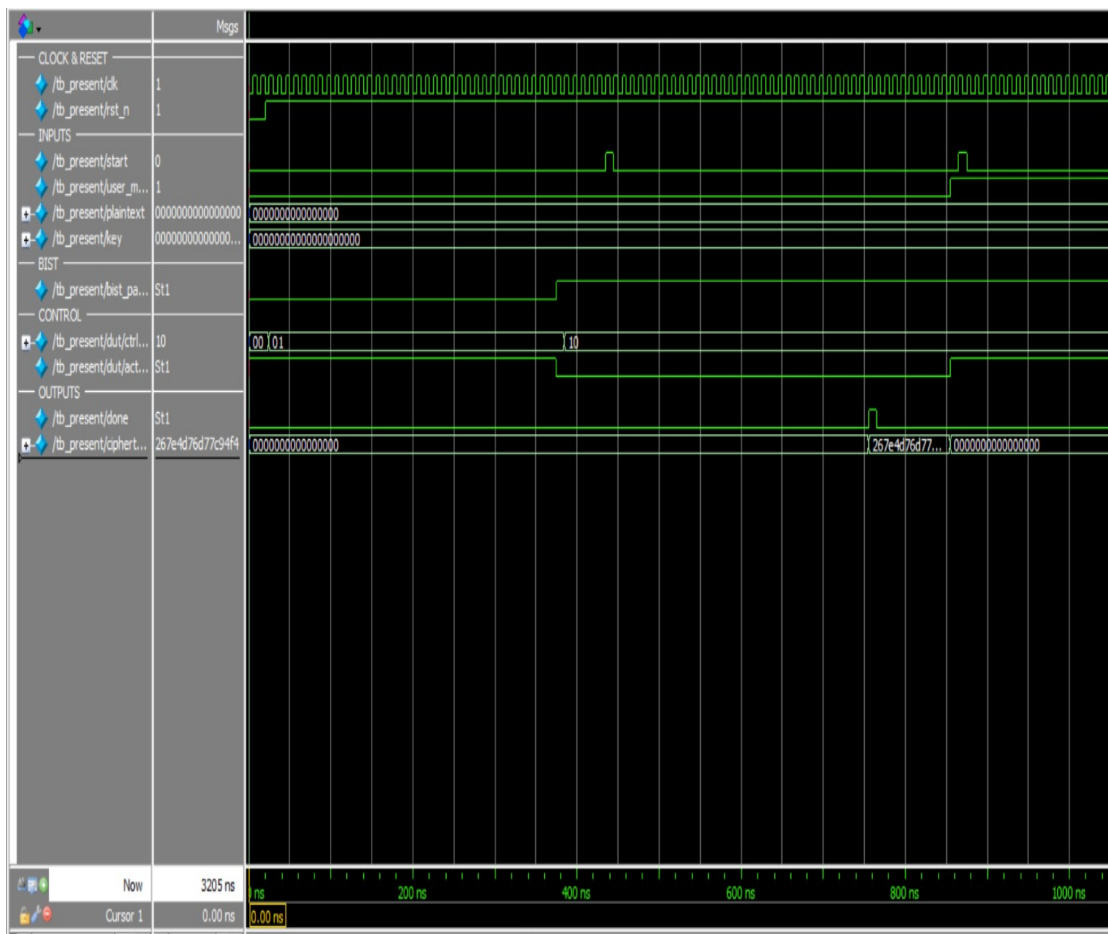
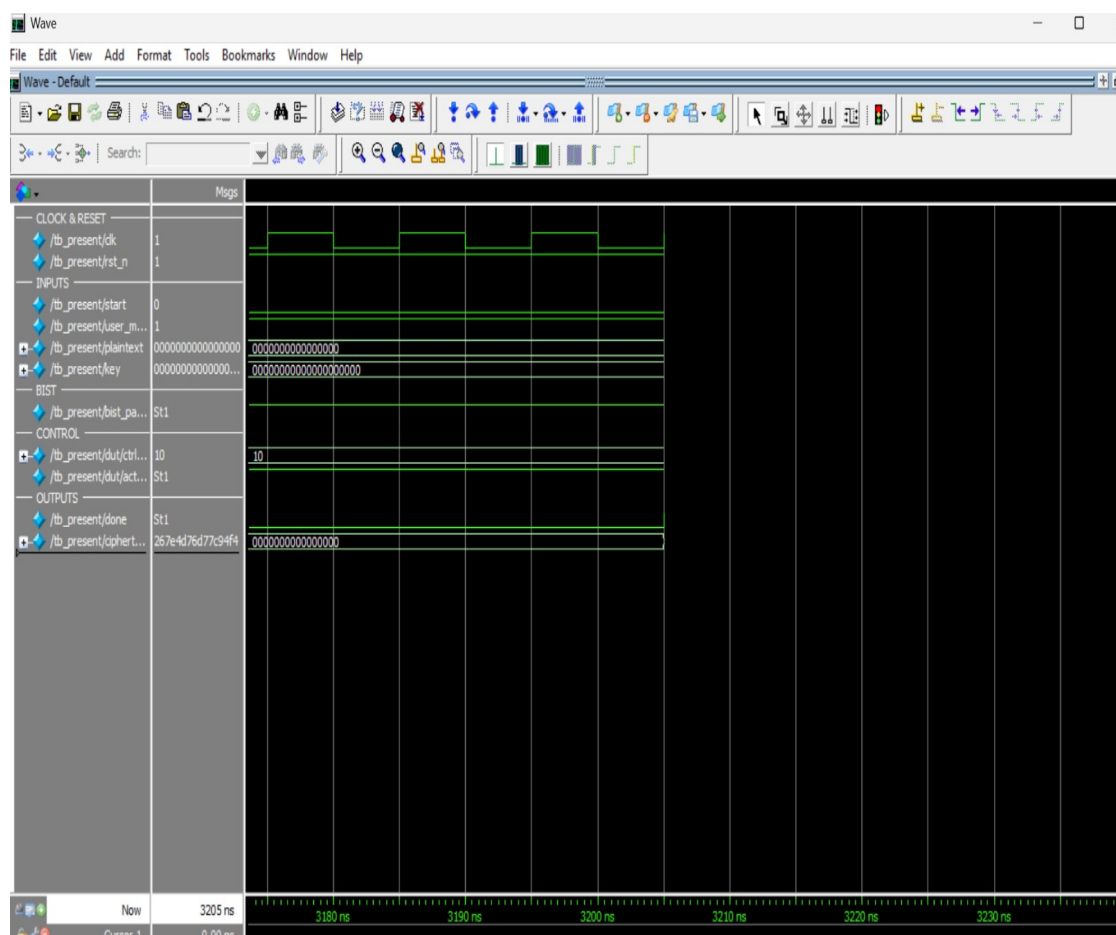


Figure 5.2: ModelSim Simulation Waveform Showing Parallel datapath

5.1.3 8-bit Serial Datapath Simulation

To evaluate the flexibility of the architecture for low-power or area-restricted scenarios, the delay-based serial behavior modeling was subjected to separate simulation, as shown in Figure 5.3. This architectural variant models low-power operation by introducing

controlled delay before processing. While the parallel mode prioritizes speed, this serial approach is specifically designed to minimize the hardware gate count and the associated dynamic switching power.



5.2 LATENCY AND PERFORMANCE ANALYSIS

The post-simulation analysis involved a comprehensive cross-verification of the ModelSim console logs and timing data to assess the efficiency and correctness of the dual-mode architecture. Figure 5.4 presents the console output generated during the complete verification phase, capturing the end-to-end operational flow from BIST initialization to final ciphertext generation in both modes.

This output confirms that both the serial and parallel modes produce mathematically identical and bit-accurate ciphertexts (0x267E4D76D77C94F4), matching the expected output of the PRESENT-80 cipher for the all-zero test vector. This agreement validates the correctness of the S-box, P-layer, and the dual S-box strategy without introducing any functional discrepancy.

The console logs further verify that internal state transitions, round counter progression, and round-key scheduling follow the PRESENT specification across all 31 rounds. Each round key evolves correctly through rotation, substitution, and counter XOR operations, ensuring cryptographic validity and zero functional deviation from the standard.

The identical ciphertext output across both modes confirms that the reconfigurable datapath maintains both security and functional accuracy. Additionally, the logs validate the Trust-Before-Performance transition, where the assertion of `bist_done` and `bist_pass` ensures safe activation of the selected datapath without any leakage of intermediate data, preserving strict data integrity during mode switching.

```

# BIST PASSED at t = 375 ns
# --- Test 1: Parallel Mode ---
# Parallel Start = 435 ns
# Parallel Done  = 755 ns
# Parallel Latency = 320 ns (32 cycles)
# Ciphertext = 267e4d76d77c94f4
# PASS: Parallel correct
# --- Test 2: Serial Mode ---
# Serial Start    = 865 ns
# Serial Done     = 3205 ns
# Serial Latency  = 2340 ns (234 cycles)
# Ciphertext = 267e4d76d77c94f4
# PASS: Serial correct
# --- Simulation Complete ---
# Serial is 7x slower than Parallel

```

Figure 5.4: Console Results

The comparative analysis reveals a clear trade-off between execution speed and hardware efficiency. As shown in Table 5.2, the Parallel mode completes encryption in 32 clock cycles (320 ns), providing the high throughput necessary for real-time secure communication. Conversely, the Serial mode requires 234 clock cycles (2340 ns), prioritizing area and power conservation over speed.

This 7x performance gap highlights the system's flexibility in addressing IoT hardware security challenges. By dynamically adapting its performance profile, the design ensures that the "Trust-Before-Performance" protocol is both secure and practically optimized for diverse field deployments.

Parameter	Parallel Mode	Serial Mode
Latency	320 ns	2340 ns
Clock Cycles	32	234
Ciphertext	0x267E4D76D77C94F4	0x267E4D76D77C94F4
BIST Status	PASS at 375 ns	—
Efficiency Ratio	1.0 (Base)	7x Latency

Table 5.1: Performance Comparison of Parallel and Serial Modes

5.3 FPGA HARDWARE IMPLEMENTATION AND VALIDATION

The design was implemented on a Xilinx Spartan-6 (XC6SLX16-FTG256, speed grade -2) FPGA using Xilinx ISE Design Suite 14.7. The implementation targeted the VPTB-20 Spartan-6 Trainer Kit operating at 20 MHz, constrained via a UCF file assigning the clock, reset, start, and LED output signals to the appropriate physical pins.

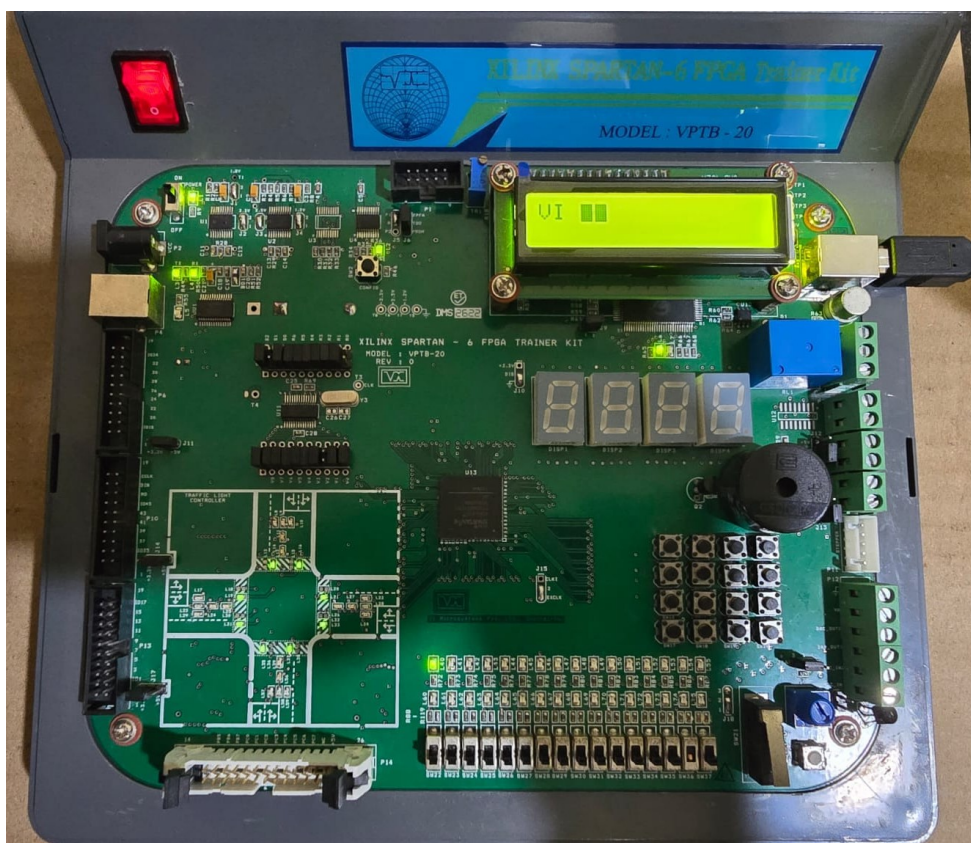


Figure 5.5: Xilinx Spartan-6 FPGA Trainer Kit (VPTB-20) used for implementation

The synthesis and implementation results are summarized below:

Parameter	Value
Device	XC6SLX16-FTG256-2
Slice Registers	592 out of 18,224 (3%)
Slice LUTs	929 out of 9,112 (10%)
Occupied Slices	290 out of 2,278 (12%)
Bonded IOBs	11 out of 186 (5%)
Clock Frequency	20 MHz
Timing Slack	43.953 ns
Maximum Achievable Frequency	166 MHz
Routing	100% complete, 0 errors

Table 5.2: FPGA Implementation Resource Utilization

The generated bitstream was programmed onto the FPGA via Xilinx iMPACT using JTAG boundary scan. Upon programming, the BIST controller executed autonomously, completing the full 31-round PRESENT encryption of the all-zero test vector. The BIST PASS LED (L40, mapped to FPGA pin A8) asserted HIGH, confirming that the hardware-computed ciphertext matched the golden reference 0x267E4D76D77C94F4 — identical to the simulation result.

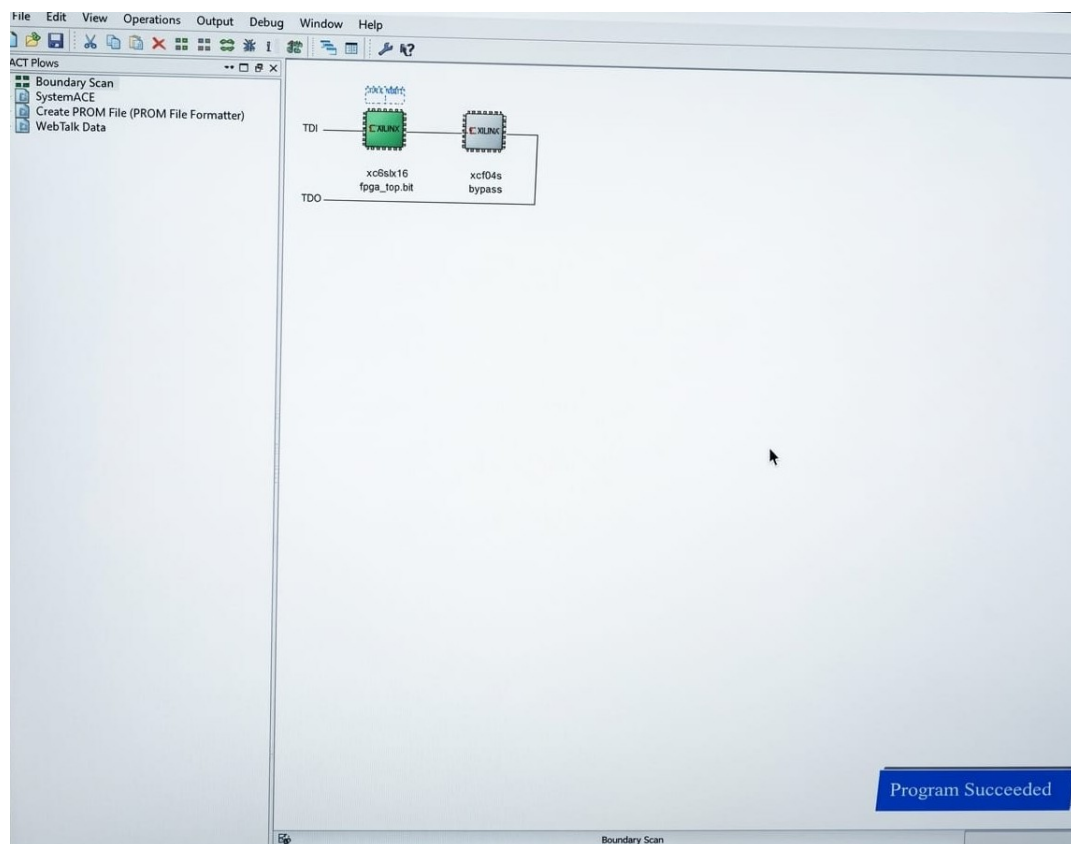


Figure 5.6: Successful FPGA Programming using Xilinx iMPACT

The Done LED (L41, pin C8) asserted following encryption completion, and the remaining LEDs (L42–L47) displayed the top 6 bits of the ciphertext output, providing a direct visual confirmation of the encrypted result on hardware.

5.4 COMPARISON WITH AES

The PRESENT lightweight block cipher is specifically designed for resource-constrained environments, whereas the Advanced Encryption Standard (AES) [5] is optimized for high-performance systems. AES is widely adopted as a global encryption standard for secure data communication. AES operates on 128-bit data blocks and uses complex operations such as MixColumns, which involve finite field arithmetic, resulting in higher computational complexity and hardware resource requirements.

In contrast, PRESENT operates on 64-bit data blocks and uses simpler substitution-permutation operations with 4-bit S-boxes and bitwise permutations. This significantly reduces hardware complexity, making it suitable for low-power applications such as IoT devices.

Feature	AES	PRESENT
Block Size	128-bit	64-bit
Key Size	128/192/256-bit	80/128-bit
Rounds	10–14	31
S-box Size	8-bit	4-bit
Complexity	High	Low
Power Consumption	High	Low
Hardware Area	Large	Small
Target Devices	Computers, Servers	IoT, Sensors

Table 5.3: Comparison between AES and PRESENT

From a hardware perspective, AES requires larger area and higher power due to its 8-bit S-boxes and matrix-based transformations, while PRESENT achieves comparable security for constrained environments with significantly fewer logic elements. However, AES provides stronger security margins due to its larger block size and more complex transformations, making it more suitable for high-security applications.

CHAPTER 6

CONCLUSION AND FUTURE WORK

The development of the Self-Tested Dual-Mode PRESENT Cipher addresses the critical need for balanced security and reliability in the evolving IoT landscape. This project has demonstrated that cryptographic cores can be designed to be both high-performing and energy-conscious without sacrificing the ability to detect internal hardware failures.

6.1 CONCLUSION

The implementation and verification of this project lead to the following conclusions as outlined in the research objectives:

- **Hardware-Efficient Design:** The architecture successfully implements the PRESENT cipher using an optimized logic footprint, making it suitable for resource-constrained IoT devices.
- **Dual-Mode Flexibility:** The system provides a versatile solution by supporting a 64-bit parallel datapath for high-speed encryption and an serial-mode-inspired datapath for diagnostic and low-power states.
- **Fault-Tolerant Reliability:** The integration of a Built-In Self-Test (BIST) module enables the system to autonomously detect hardware faults, ensuring the integrity of the encryption process in long-lifetime devices.
- **Verified Accuracy:** Simulation results in ModelSim confirmed that the design is 100% mathematically accurate, achieving the correct ciphertext within the predicted timing constraints (32 cycles for parallel and 234 cycles for serial).

6.2 FUTURE WORK

While the current work establishes a verified RTL model, the following milestones are identified for future expansion of this project:

- **Extended FPGA Analysis:** The design has been successfully validated on the Xilinx Spartan-6 FPGA, confirming correct hardware operation at 20 MHz with 10% LUT utilization. Future work includes generating detailed Power, Area, and Delay (PAD) reports and implementing the design on more constrained FPGA families such as the Artix-7 to further evaluate physical performance metrics for ultra-low-power IoT deployment.
- **Area and Power Optimization:** Further architectural refinement will be conducted to generate Power, Area, and Delay (PAD) reports, aiming to minimize the silicon overhead for ultra-low-power applications.
- **Integration with IoT Communication Modules:** Future work includes interfacing the cryptographic core with physical IoT protocols such as Zigbee or LoRa to secure real-world data transmissions.
- **Hardware Acceleration for Embedded Systems:** The design can be expanded as a standalone cryptographic co-processor to provide hardware acceleration for various embedded microcontrollers.

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